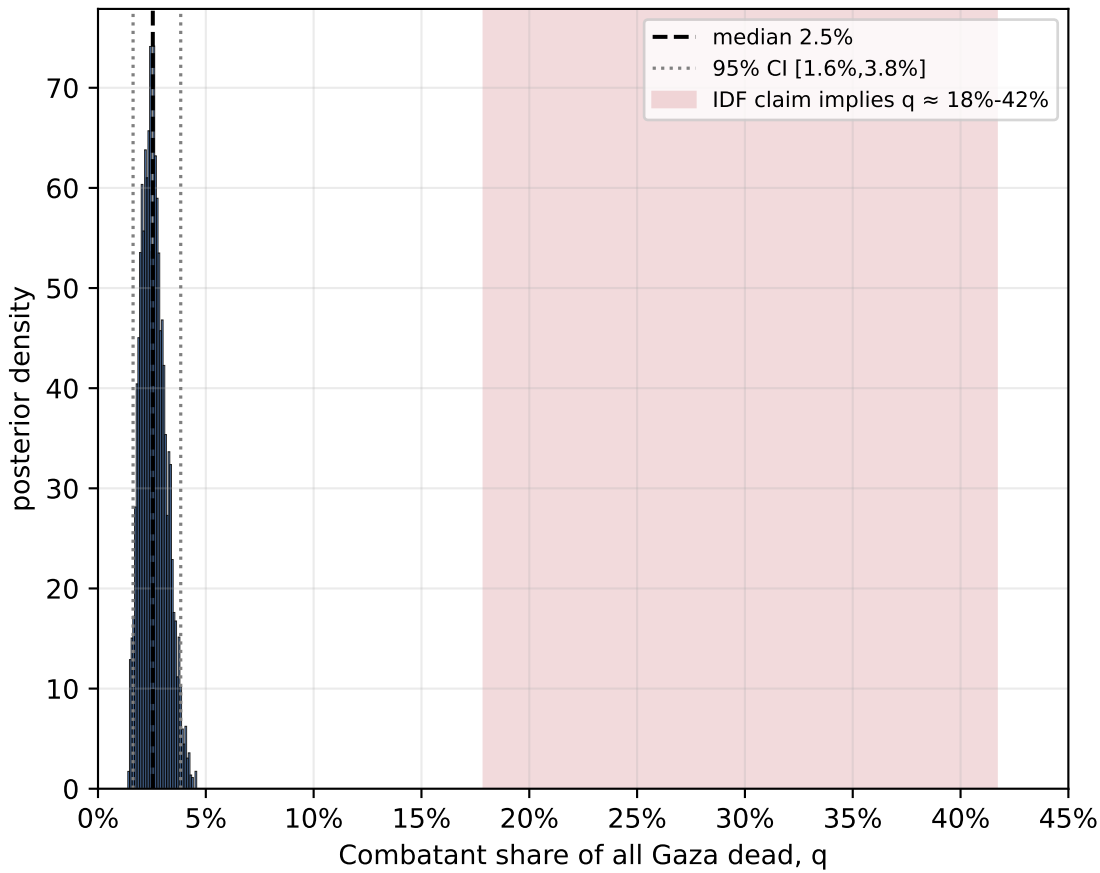


Gaza war — spatial Bayesian inference of militant vs civilian death share

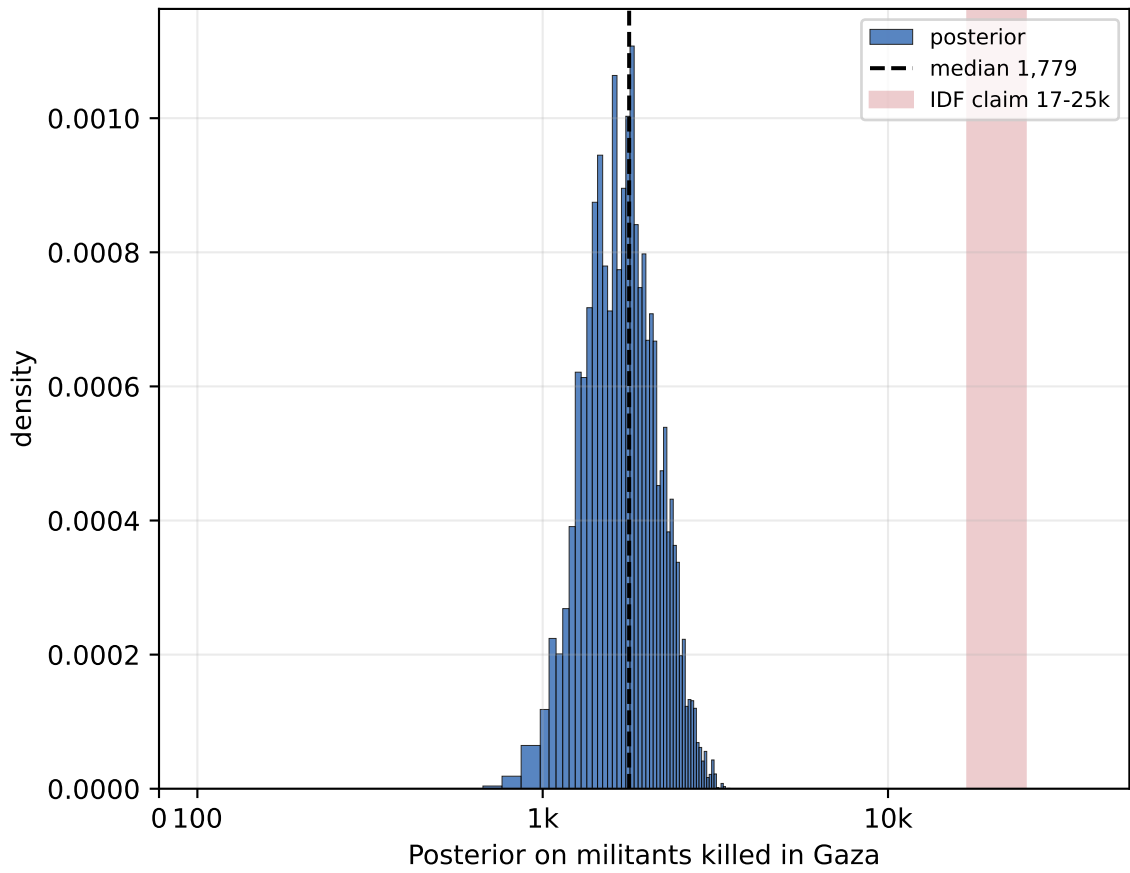
Forward model: 5 governorates × 80 cells; latent militant cell-density with clustering γ ; strikes target high-militant cells with bias α ; civilian adult-male exposure μ_M absorbs male over-representation.

Conditioned on PCBS demographics + MoH+OHCHR demographic split + MoH death tally + Lancet undercount. No belligerent claims used as priors.

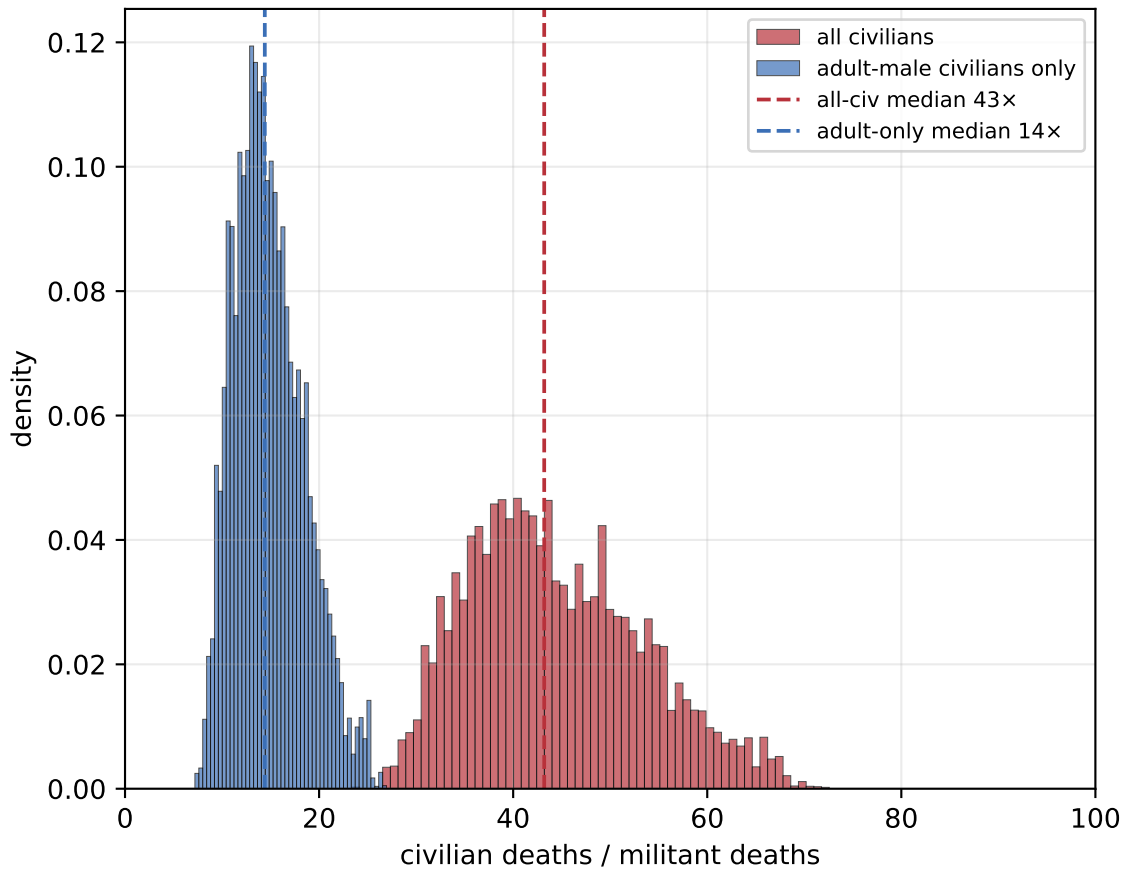
(1) Posterior on militant share q
median 2.5%, 95% CI [1.6%, 3.8%]



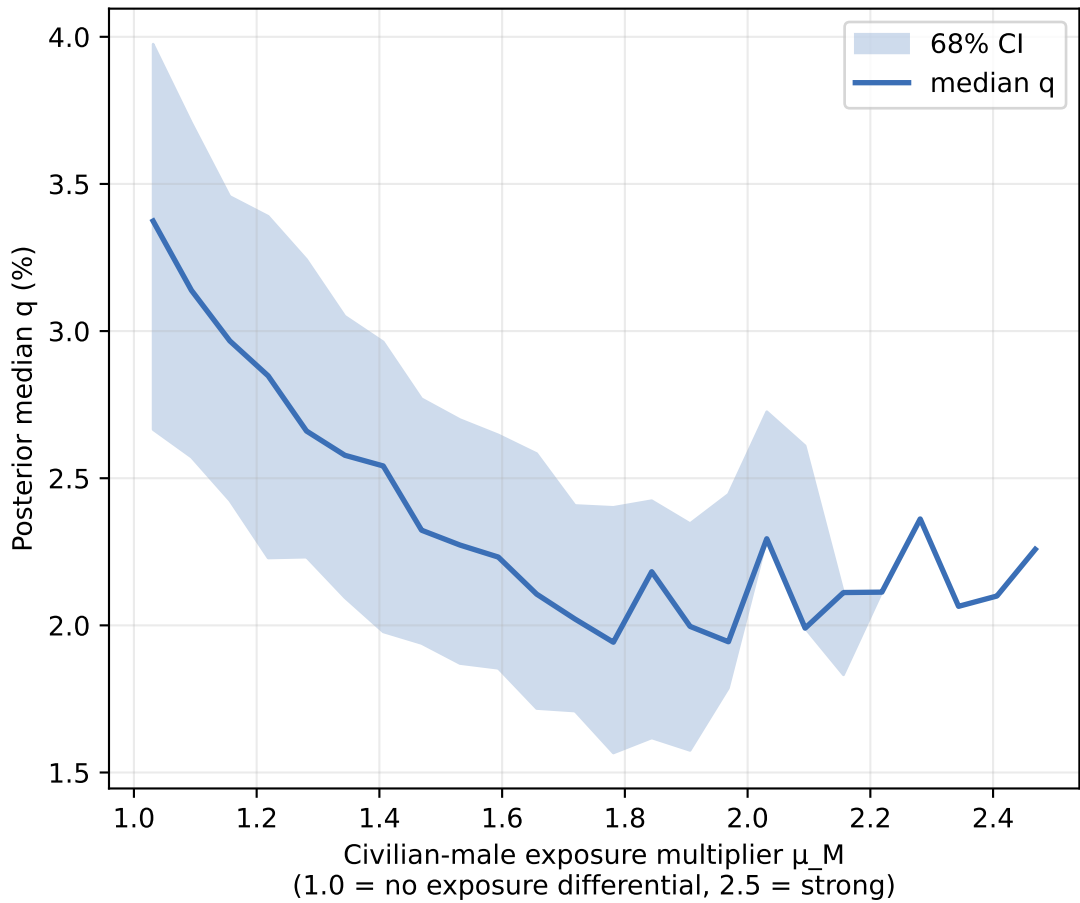
(2) Implied militants killed: median 1,779
95% CI [1,112, 2,720]



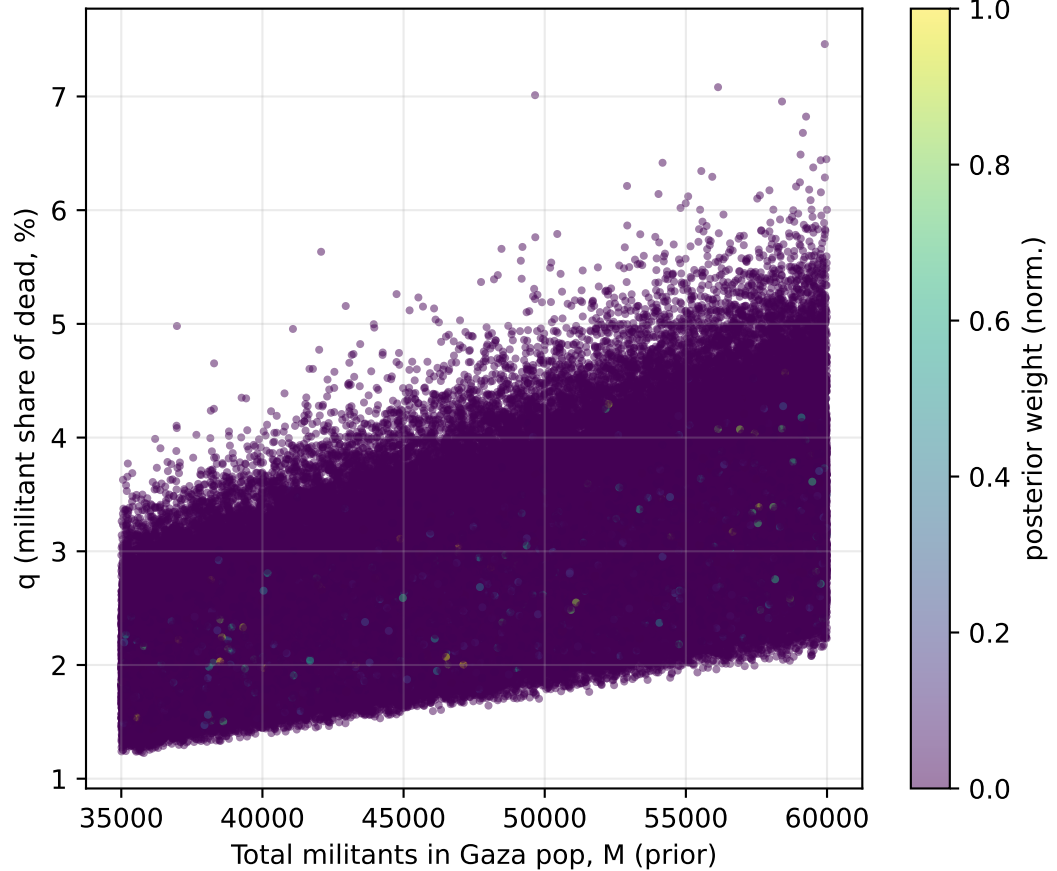
(3) Posterior on civ:mil ratio



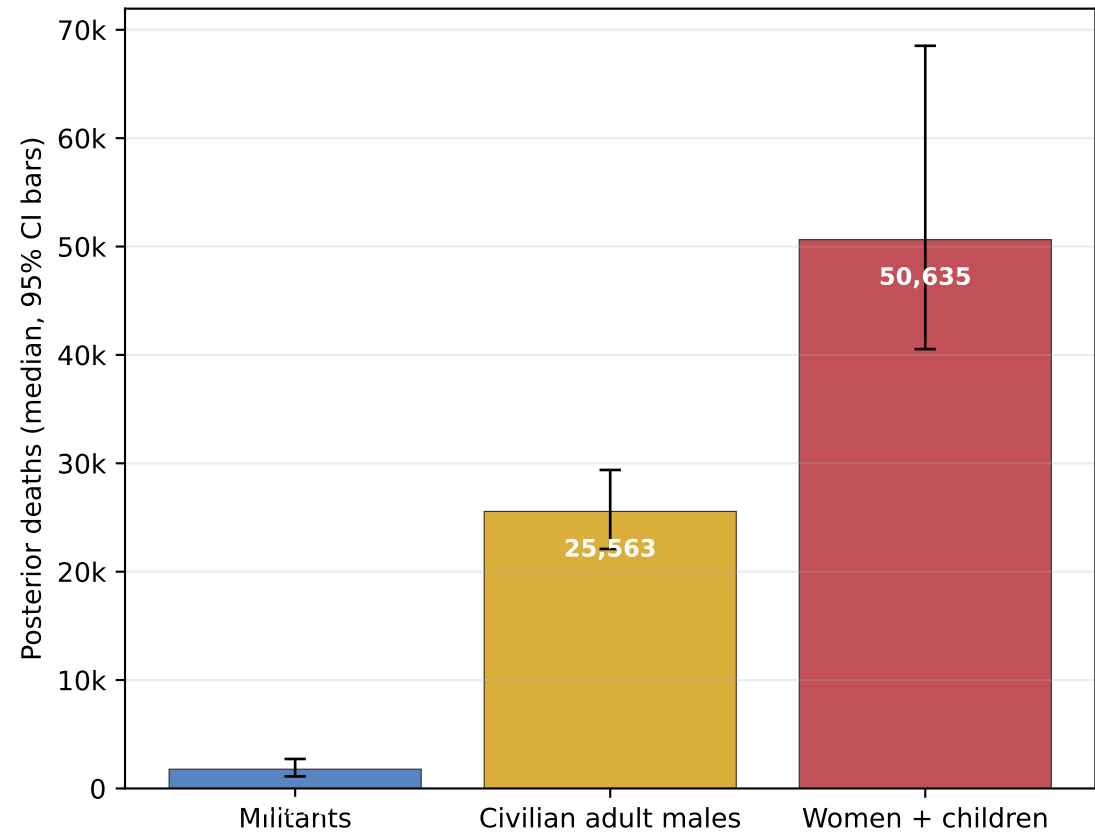
(4) Sensitivity: militant share q vs μ_M
(if civilian men were *not* over-exposed, q would rise)



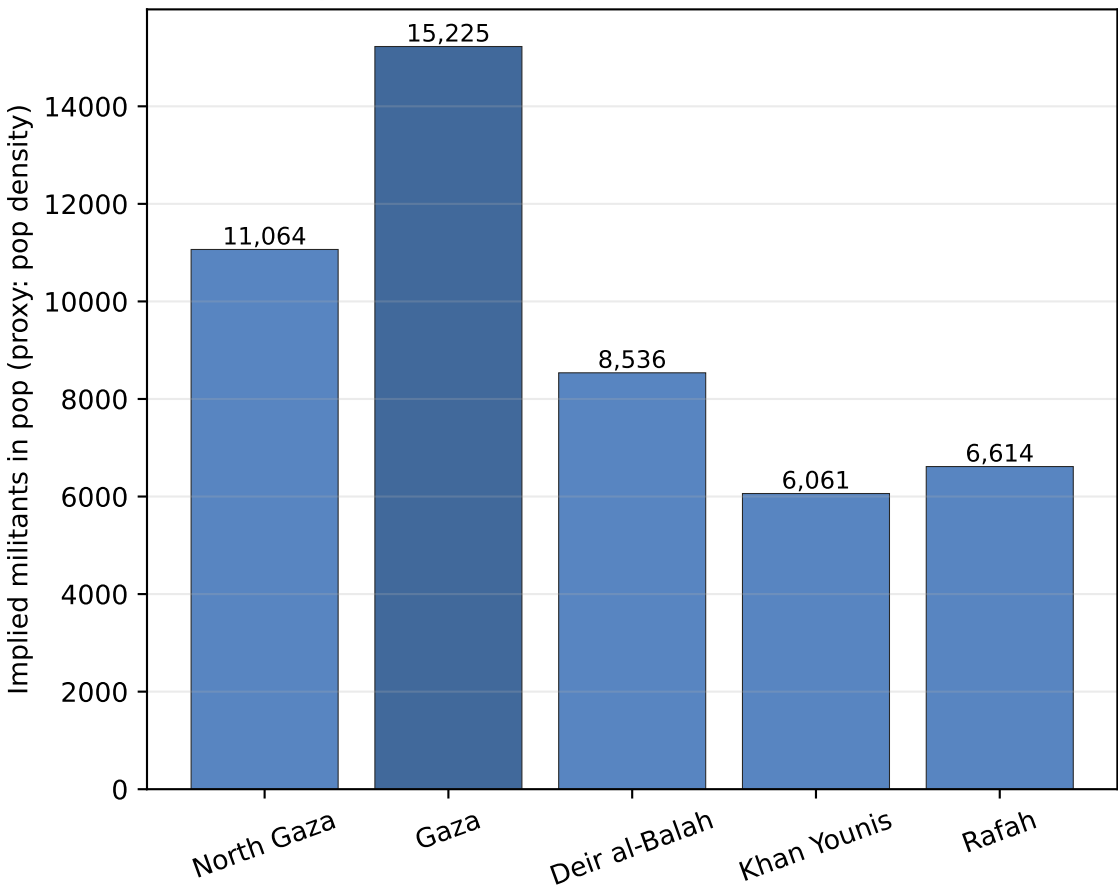
(5) Joint posterior over (M , q)



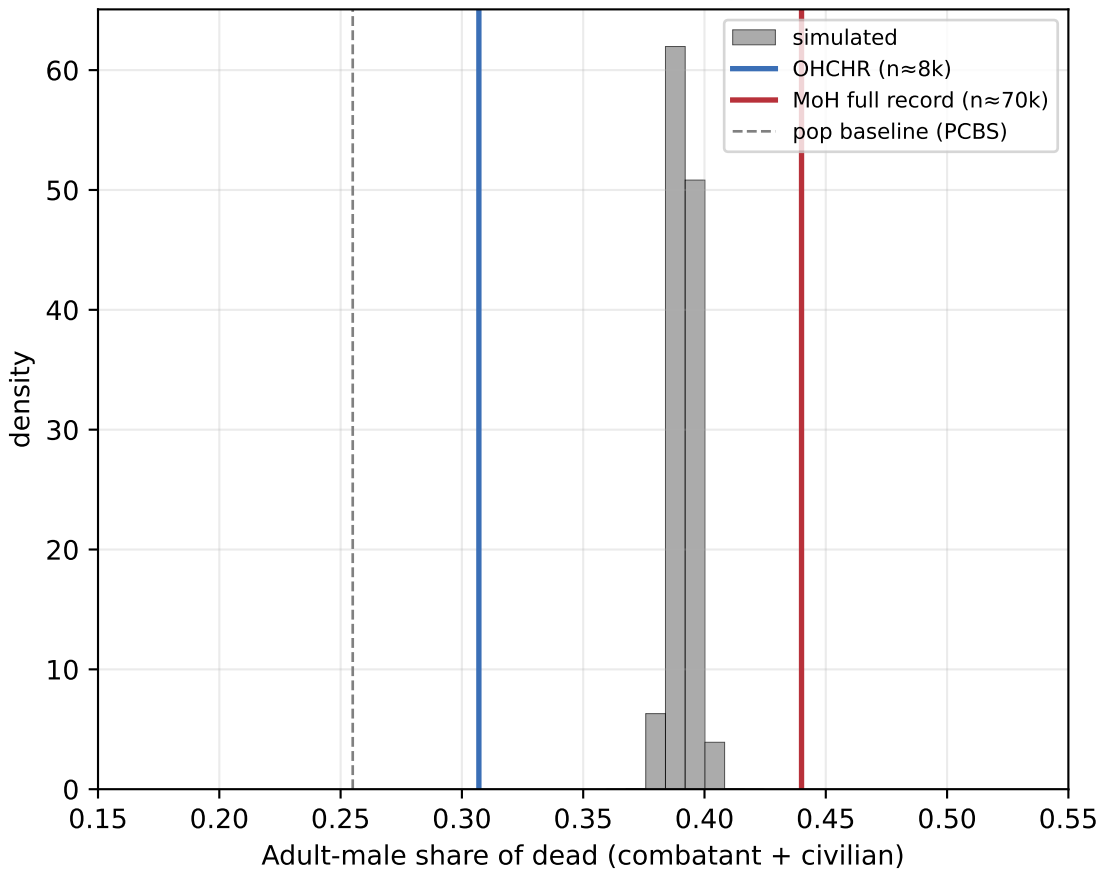
(6) Posterior decomposition of 70,100 reported deaths



(7) Implied militants by governorate (sum=47,503)



(8) Posterior fit to demographic data



Posterior (importance sampler, ESS=2,277)

Direct deaths in Gaza (true):
D_{total} median 78,298, 95% CI [65,384, 97,441]

Decomposition (median):
Militants killed 1,779
Adult-male civilians 25,563
Women + children 50,635

Militant share q median 2.5%
95% CI [1.6%, 3.8%]

Civilian:militant ratio (incl. kids):
median 43x, 95% CI [30x, 64x]

For comparison:
IDF/Israeli intel claim 17–25k militants killed
⇒ $q \approx 19\text{--}42\%$ (incompatible with posterior)
Naïve OHCHR-based Bayes $q \approx 6.0\%$ (CI 2.4–9.2%)
Simulator (this fig.) $q \approx 2.5\%$ (CI 1.6–3.8%)